Local Compactness and Global Repulsion: A Memory-Queue Framework for Imbalanced Skin Lesion Classification

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ABSTRACT

Class imbalance in skin lesion classification severely skews the learned embedding space, leaving minority-class representations sparse, dispersed, and easily confused with majority clusters. We address this with a memory queue-guided dual-scale metric learning framework evaluated on the ISIC 2018 and ISIC 2019 benchmarks. Our approach pairs a Weighted K-Nearest Neighbours loss for local neighbourhood compactness with a heavy-tailed Cauchy Prototype loss for global class separation, both computed over a dynamically maintained Exponential Moving Average (EMA) memory queue that gives minority classes a persistent, stable similarity pool despite mini-batch sampling sparsity. The method achieves 87.41% ± 2.03% overall accuracy and 79.32% ± 1.87% balanced accuracy on ISIC 2018, and 82.78% ± 1.97% overall accuracy and 71.97% ± 1.93% balanced accuracy on ISIC 2019, outperforming PaCo and other loss-function baselines on balanced accuracy and most evaluation and embedding metrics. Quantitative representation analysis confirms these gains reflect genuine geometric restructuring rather than reweighting alone, yielding higher KNN purity, wider class margins, and stronger global separation. Source code is available at https://github.com/mukut02/icvgip_skin_cancer.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; **Supervised learning**; • **Applied computing** → *Health informatics*.

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KEYWORDS

Skin Cancer, Class Imbalance, K-Nearest Neighbours, Cauchy Kernel, Prototype Learning, Memory Queue

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1 INTRODUCTION

Early and accurate diagnosis of skin cancer significantly improves patient outcomes, yet it remains a visually demanding task even for experienced clinicians. Dermoscopy-based automated classification has emerged as a promising decision-support tool, with large-scale benchmarks such as ISIC 2018 [1, 2] and ISIC 2019 [2–4] driving rapid progress using deep learning-based methods. These datasets, however, reflect the epidemiological reality of skin disease: common benign conditions such as melanocytic nevi vastly outnumber clinically critical but rare categories such as dermatofibroma and vascular lesions. This long-tailed distribution puts standard classifiers in a bind, since maximising overall accuracy usually means suppressing minority categories, which are often the very lesion types where a missed or wrong diagnosis matters most. As a result, standard classifiers achieve misleadingly high overall accuracy while suffering from abysmal sensitivity on rare categories—creating severe clinical risks where life-threatening or diagnostic-critical lesions are systematically misclassified as benign nevi.

Contributions: We propose a multi-objective formulation that uses weighted cross-entropy with two EMA memory queue [5, 6] guided geometric losses, each operating at a different scale. Concretely, we contribute the following:

(1) A **Weighted K-Nearest Neighbours (KNN) loss** [7, 8] that enforces local neighbourhood compactness by contrasting top-k same-class neighbours against the hardest inter-class

negative in the queue, with class-rebalanced sample weighting applied to tighten intra-class clusters and resolve local decision boundary ambiguities.

(2) A **Cauchy Prototype Separation loss** [9] that estimates a running class prototype from the queue for every class and pushes these prototypes apart via a heavy-tailed kernel, whose gradients stay non-vanishing even once classes are moderately well separated, so global class separation keeps improving well into training rather than plateauing early.

Our ablation and embedding-quality analysis confirms that the two objectives tackle distinct, complementary failure modes of imbalanced training.

2 RELATED WORK

Reweighting and margin-based losses: A common strategy for long-tailed recognition rebalances the classifier head rather than the feature extractor: effective-number weighting [10], LDAM [11], Balanced Softmax [12], and Focal Loss [13] reweight or rescale the loss by class frequency to curb majority bias. These methods operate purely at the logit level, leaving the embedding space unconstrained—nothing compels the encoder to form compact, well-separated representations for rare classes. Minority-class embeddings can thus remain dispersed and entangled with majority clusters even after reweighting, a gap our ablation study directly probes.

Representation learning for class imbalance: A complementary line of work targets feature-space geometry directly. SupCon [14] shows supervised contrastive objectives yield more discriminative embeddings than cross-entropy alone, while PaCo [15] extends this with learnable prototypes anchoring minority classes. Center Loss [16], ArcFace [17], and Prototypical Networks [9] similarly treat intra-class compactness and inter-class separation as explicit objectives. A shared limitation is that SupCon and PaCo compute objectives over batch-level pairs or batch-visible prototypes: under long-tailed sampling, minority classes appear infrequently—or not at all—within a batch, starving these objectives of signal where it is needed most.

Memory queues for stable representation learning: Maintaining a bank of feature embeddings decoupled from batch size traces back to Wu et al. [6], who stored one representation per instance for non-parametric instance discrimination. MoCo [5] extends this with a momentum-updated queue, building on the EMA principle for stable target representations [18], decoupling negative-sample diversity from batch size. We adapt this to the supervised, class-imbalanced setting via EMA updates on per-class queue entries, so every class—regardless of batch frequency—maintains a stable, representative embedding pool, letting our local (KNN) and global (prototype) objectives draw consistent signal for minority classes throughout training.

3 METHODOLOGY

Given training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, an encoder f_θ produces a feature embedding $z_i = f_\theta(x_i)$, which is ℓ_2-normalised to map it onto the unit hypersphere:

$$\tilde{z}_i = \frac{z_i}{\|z_i\|_2 + \epsilon}. \tag{1}$$

The full objective is defined as:

$$\mathcal{L} = \mathcal{L}_{WCE} + \lambda_{knn}\mathcal{L}_{knn} + \lambda_{proto}\mathcal{L}_{proto}. \tag{2}$$

We present a multi-objective metric learning framework to tackle severe class imbalance in skin lesion classification via an EMA memory queue. The methodology combines Weighted Cross-Entropy loss $\mathcal{L}_{WCE}$ with a weighted KNN loss $\mathcal{L}_{knn}$ (weight λ_{knn}) for local intra-class compactness, and a Cauchy Prototype Separation loss $\mathcal{L}_{proto}$ (weight λ_{proto}) for global inter-class repulsion. By operating on stable queue statistics rather than skewed mini-batches, this dual-scale approach regularises the embedding space's geometry to prevent dispersion and entanglement of minority classes. Neither objective is sufficient on its own: local compactness alone can still leave the class as a whole crowded against a neighbouring class, and global separation alone provides no gradient signal for hard, easily-confused individual samples. The diagram of the model is shown in Figure 1.

Class-Rebalanced Classification loss: To reduce class frequency bias, we apply effective-number weighting [10] rather than naive inverse-frequency weighting [19]. For class c with n_c training samples and $\beta \in [0, 1)$ controlling sample redundancy, the per-class weight w_c is defined as:

$$w_c = \frac{1-\beta}{1-\beta^{n_c}}. \tag{3}$$

The classification objective is then:

$$\mathcal{L}_{WCE} = -\frac{1}{B}\sum_{i=1}^{B} w_{y_i} \log p(y_i|x_i), \tag{4}$$

where B is the batch size, and $p(y_i|x_i)$ denotes the predicted softmax probability assigned to the correct ground-truth class y_i for sample x_i.

EMA Memory Queue: A central difficulty with metric-learning losses such as $\mathcal{L}_{knn}$ and $\mathcal{L}_{proto}$ is that they require a rich, diverse pool of same-class and cross-class embeddings to compute meaningful positive and negative similarities - a requirement that a single mini-batch, especially under severe class imbalance, cannot satisfy. With batch size B typically in the low hundreds and highly skewed class frequencies, minority classes may be represented by only one or two samples per batch (or none at all), making in-batch positive mining unstable and in-batch prototype estimates noisy. The EMA memory queue addresses this by decoupling the pool of candidate embeddings from the current batch: it maintains a much larger, persistent bank of $M \gg B$ embeddings spanning all classes, so that every batch — regardless of which classes happen to appear in it — has access to a stable and diverse set of positives, hard negatives, and per-class prototypes.

We dynamically maintain an Exponential Moving Average (EMA) memory queue $Q = \{(q_j, \ell_j)\}_{j=1}^{M}$ of ℓ_2-normalised 512-dimensional embeddings, with q_j denoting the j-th stored embedding and ℓ_j its corresponding class label. On each forward pass, given a new anchor embedding $\tilde{z}_i$ with label y_i, we update the queue entry at position p as:

$$(q_p, \ell_p) \leftarrow \begin{cases} (\gamma\, q_p + (1-\gamma)\tilde{z}_i, \ell_p), & \ell_p = y_i, \\ (\tilde{z}_i, y_i), & \text{otherwise.} \end{cases} \tag{5}$$

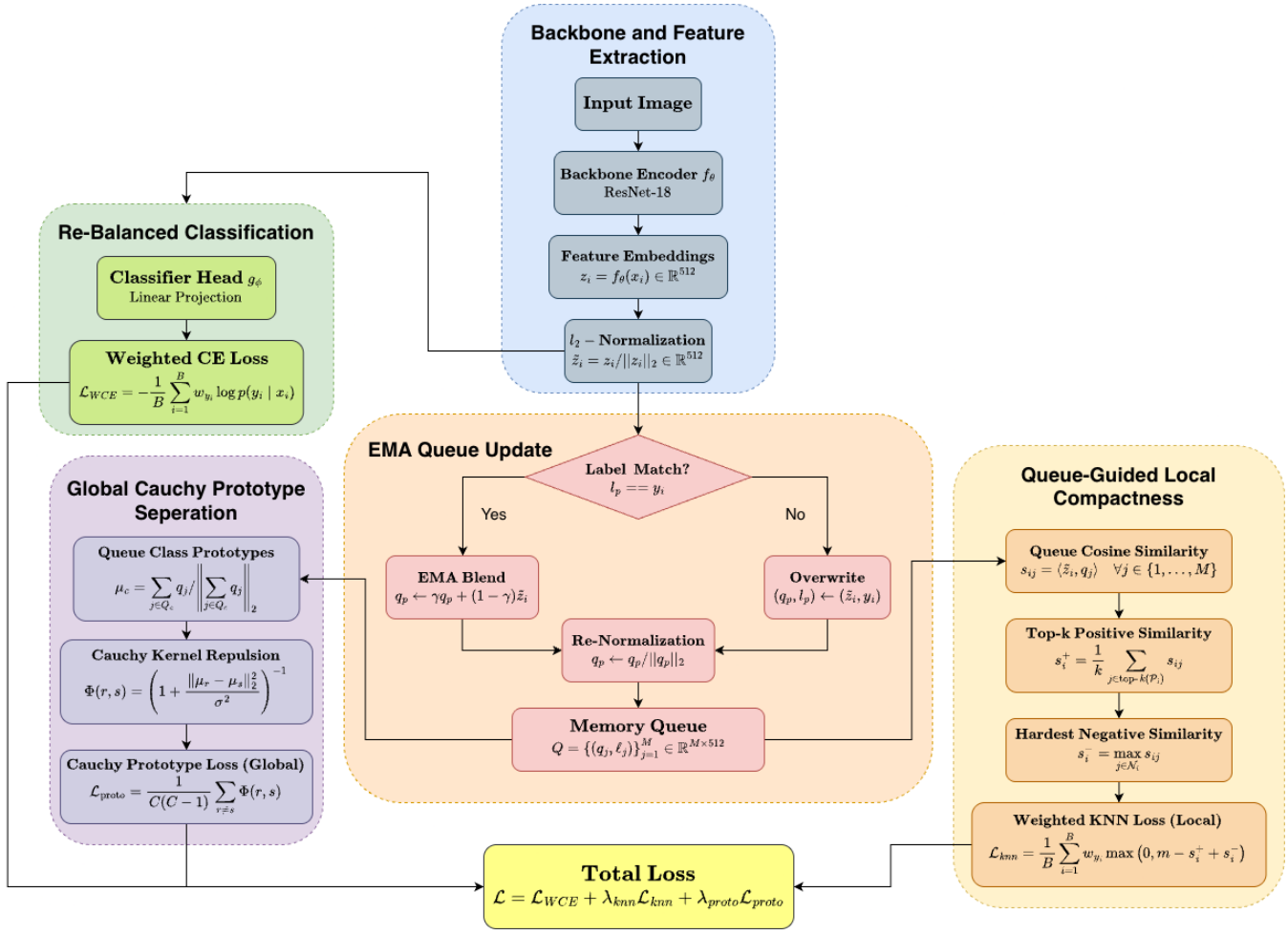


Figure 1: Workflow of the proposed model

To ensure the updated embedding remains on the unit hypersphere, we re-normalise:

$$q_p \leftarrow \frac{q_p}{\|q_p\|_2}. \tag{6}$$

Here, $\gamma \in [0, 1)$ is the EMA momentum coefficient. The EMA rule smoothly blends historical representations for the matching class, preventing stale entries while preserving intra-class diversity. While the queue acts as a sliding window over the data distribution and maintains the overall dataset imbalance, its large capacity ($M \gg B$) ensures that minority classes are adequately represented with a stable, slowly-evolving set of embeddings. In effect, both $\mathcal{L}_{knn}$ and $\mathcal{L}_{proto}$ can query against this rich pool, giving both losses a far more reliable statistical basis than the current mini-batch alone could provide.

Weighted KNN Loss: The motivation for $\mathcal{L}_{knn}$ is to enforce *local* discriminability using two targeted strategies:

- **Top-k Positives:** We average the top-k positive similarities, providing a robust anchor to the local class neighbourhood that is less sensitive to outliers than a single nearest positive.
- **Hardest Negative:** We penalise only the single most confusable inter-class embedding. This directly pushes apart visually similar lesion subtypes at the decision boundary where misclassifications actually occur.

Similarities s_{ij} between an anchor $\tilde{z}_i$ and queue entries are computed via dot product: $s_{ij} = \langle \tilde{z}_i, q_j \rangle$. We define the positive set $P_i = \{j : \ell_j = y_i\}$ and the negative set $N_i = \{j : \ell_j \neq y_i\}$. The positive similarity s_i^+ is averaged over the top-k nearest neighbours, while the negative similarity s_i^- targets the hardest inter-class negative:

$$s_i^+ = \frac{1}{k} \sum_{j\in \text{top-}k(P_i)} s_{ij}, \qquad s_i^- = \max_{j\in N_i} s_{ij}. \tag{7}$$

The neighbourhood loss $\mathcal{L}_{knn}^i$ for anchor i enforces a margin m:

$$\mathcal{L}_{knn}^i = \max\left(0,\ m - s_i^+ + s_i^-\right). \tag{8}$$

Because the positive and negative sets are drawn from the memory queue Q rather than the current batch, P_i and N_i remain well-populated even for rare classes, so the top-k average and hardest-negative search are computed over a genuinely representative sample rather than the few (or zero) in-batch instances of a minority

class. Class-balanced weighting is applied per sample to further emphasise minority classes, ensuring the margin violation for an under-represented class contributes proportionally more to the gradient than an easy majority-class violation:

$$\mathcal{L}_{knn} = \frac{1}{B}\sum_{i=1}^{B} w_{y_i}\mathcal{L}_{knn}^{i}. \tag{9}$$

Cauchy Prototype Separation Loss: While $\mathcal{L}_{knn}$ shapes fine-grained, sample-level neighbourhoods, it provides no explicit guarantee that entire classes are well-separated at the aggregate level — two classes could remain locally compact yet have their overall centroids drift close together, which is a particularly common failure mode for visually similar minority classes with limited training signal. $\mathcal{L}_{proto}$ is introduced to directly address this at the level of class prototypes rather than individual samples, complementing the local objective with a global one.

Estimating each prototype from the current mini-batch alone would be unreliable under imbalance, since a batch may contain few or no samples of a given class. Estimating it from the queue alone would detach it from the encoder. We therefore form each prototype as a *joint batch–queue centroid*. For class c, let $\mathcal{B}_c$ be the batch samples and Q_c the queue entries of that class:

$$v_c = \sum_{i\in\mathcal{B}_c}\tilde{z}_i + \sum_{j\in Q_c}\mathrm{sg}(q_j), \qquad \mu_c = \frac{v_c}{\|v_c\|_2+\epsilon}, \tag{10}$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient. Queue entries are detached, but batch embeddings $\tilde{z}_i(\theta)$ keep their autograd connection to the encoder. For a batch sample i of class r and any other class s, the gradient of $\mathcal{L}_{proto}$ is therefore non-zero:

$$\frac{\partial\mathcal{L}_{proto}}{\partial\tilde{z}_i} \;\propto\; \frac{(\mathbf{I}-\mu_r\mu_r^{\top})(\mu_r-\mu_s)}{\|v_r\|_2} \neq \mathbf{0}. \tag{11}$$

where Q_c denotes queue entries belonging to class c. For two class prototypes μ_r and μ_s, the squared distance d_{rs}^2 is

$$d_{rs}^2 = \|\mu_r-\mu_s\|_2^2.$$

Pairwise similarity is measured using a Cauchy kernel ($\Phi(r,s)$), where σ is the kernel bandwidth — a fixed hyperparameter controlling how far apart prototypes must be before the Cauchy kernel's repulsion effectively fades out.

$$\Phi(r,s) = \frac{1}{1+d_{rs}^2/\sigma^2}.$$

Thus, the prototype objective ($\mathcal{L}_{proto}$) is:

$$\mathcal{L}_{proto} = \frac{1}{C(C-1)}\sum_{r\neq s}\Phi(r,s),$$

Unlike a Gaussian kernel [20], whose gradients vanish once classes are moderately separated, the heavy-tailed Cauchy kernel [21] sustains non-vanishing repulsion throughout training, continuously driving class centroids toward a uniformly separated arrangement on the unit hypersphere. This is especially important for minority classes: once their prototype drifts a moderate distance from its neighbours, a Gaussian-style penalty would stop providing meaningful gradient, allowing the class to settle into a suboptimal, only-partially-separated position; the Cauchy kernel's heavier tail keeps exerting a (diminishing but non-zero) repulsive force, continuing to push rare-class prototypes toward better global separation even late in training. This global, prototype-level repulsion complements the local compactness enforced by the KNN objective, so that minority classes are simultaneously coherent internally (via $\mathcal{L}_{knn}$) and well-separated from all other classes at the aggregate level (via $\mathcal{L}_{proto}$).

EXPERIMENTAL SETUP

Software and Hardware Details. All experiments were run on Kaggle's cloud infrastructure, using a single NVIDIA Tesla T4 GPU (16 GB GDDR6, 2560 CUDA cores, 320 Turing-architecture Tensor Cores). The model was built in PyTorch, with a pretrained ResNet-18 backbone serving as the feature encoder, fine-tuned end-to-end throughout training. Dataset splitting and evaluation were handled with Scikit-learn, while Matplotlib and Seaborn were used to generate plots and visualize results. NumPy and Pandas supported numerical and tabular data handling, and OpenCV was used for image preprocessing and augmentation.

Datasets: For evaluation, we use two highly imbalanced skin lesion datasets: ISIC 2018 and ISIC 2019. To ensure robust model selection and evaluation, both datasets are partitioned using a stratified 80/10/10 split for training, validation, and testing, respectively.
ISIC 2018 contains 10,015 dermoscopic images across seven lesion classes (MEL, NV, BCC, AKIEC, BKL, DF, VASC). The dataset is heavily long-tailed: NV accounts for roughly 67% of samples, while minority classes such as DF and VASC contain fewer than 150 images each. Under our stratified split, this yields 8,012 training, 1,001 validation, and 1,002 test images.
ISIC 2019 extends the benchmark to 25,331 images across eight classes (MEL, NV, BCC, AK, BKL, DF, VASC, SCC). The imbalance is comparably severe: NV dominates with over 12,000 samples, while DF and VASC each contribute fewer than 300 samples. The stratified split yields 20,264 training, 2,533 validation, and 2,534 test images.

Implementation Details: All experiments use a pretrained ResNet-18 backbone trained end-to-end on 224×224 images normalised with ImageNet statistics. Augmentations include random resized crop, flip, $\pm20^\circ$ rotation, mild colour jitter, CLAHE contrast normalization, and random erasing. We use AdamW (lr $= 3\times10^{-4}$, weight decay $= 5\times10^{-4}$) with cosine annealing over 60 epochs and batch size 128.

Loss Weights and Hyperparameters: Both λ_{knn} and λ_{proto} follow a warm-up schedule to avoid destabilising the encoder before the queue reaches sufficient coverage as defined by $\lambda(t)$:

$$\lambda(t) = \lambda_{\max}\frac{1-e^{-t/\tau}}{1-e^{-T/\tau}}, \tag{12}$$

with epochs $T = 60$, temperature $\tau = 15$, $\lambda_{knn}^{\max} = \lambda_{proto}^{\max} = 0.5$. The batch size B here is 128 and β is 0.999. The KNN loss uses neighbourhood size $k = 10$ and margin $m = 0.2$; the Cauchy kernel bandwidth is $\sigma = 0.5$. The queue stores $M = 4096$ embeddings of dimension 512, updated with EMA $\gamma = 0.99$. The hyperparameters were determined using standard hyperparameter tuning methods on validation set.

Algorithm 1 Queue-Guided Metric Learning Framework

Require: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$; encoder f_θ; classifier g_ϕ; memory queue $Q = \{(q_j, \ell_j)\}_{j=1}^{M}$; parameters $k, m, \sigma, \gamma, \beta, \tau$; max loss weights $\lambda_{knn}^{\max}, \lambda_{proto}^{\max}$.

Initialize: Pre-compute effective-number weights $w_c = (1-\beta)/(1-\beta^{n_c})$ for $c \in \{1, \ldots, C\}$

Initialize: Populate memory queue Q with normalized features from random samples; set pointer $p \leftarrow 1$

for epoch $t = 1 \ldots T$ **do**

  Compute scheduled loss weights: $\lambda_a(t) \leftarrow \lambda_a^{\max} \cdot \frac{1-e^{-t/\tau}}{1-e^{-T/\tau}}$ for $a \in \{\text{knn}, \text{proto}\}$

  for mini-batch $\{(x_i, y_i)\}_{i=1}^{B} \subset \mathcal{D}$ **do**

    Step 1: Forward Pass & Feature Normalization

    for $i = 1 \ldots B$ **do**

      $z_i \leftarrow f_\theta(x_i), \quad \tilde{z}_i \leftarrow \frac{z_i}{\|z_i\|_2 + \epsilon}, \quad a_i \leftarrow g_\phi(\tilde{z}_i)$

    end for

    Step 2: Class-Rebalanced Classification Loss

    $\mathcal{L}_{\text{WCE}} \leftarrow -\frac{1}{B} \sum_{i=1}^{B} w_{y_i} \log \text{softmax}(a_i)_{y_i}$

    Step 3: Weighted KNN Loss

    for $i = 1 \ldots B$ **do** ▷ Cross-batch search over persistent queue Q

      $s_{ij} \leftarrow \langle \tilde{z}_i, q_j \rangle \quad \forall j \in \{1, \ldots, M\}$

      Partition queue: $P_i \leftarrow \{j \mid \ell_j = y_i\}, \quad N_i \leftarrow \{j \mid \ell_j \neq y_i\}$

      if $|P_i| > 0$ **and** $|N_i| > 0$ **then**

        $k_i \leftarrow \max(1, \min(k, |P_i|)), \quad s_i^+ \leftarrow \frac{1}{k_i} \sum_{j \in \text{top-}k_i(P_i)} s_{ij}, \quad s_i^- \leftarrow \max_{j \in N_i} s_{ij}$

        $\mathcal{L}_{\text{knn}}^i \leftarrow \max(0,\ m - s_i^+ + s_i^-)$

      else

        $\mathcal{L}_{\text{knn}}^i \leftarrow 0$

      end if

    end for

    $\mathcal{L}_{\text{knn}} \leftarrow \frac{1}{B} \sum_{i=1}^{B} w_{y_i} \mathcal{L}_{\text{knn}}^i$

    Step 4: Cauchy Prototype Loss (Global Repulsion)

    for class $c = 1 \ldots C$ **do** ▷ Joint batch–queue centroids

      $\mathcal{B}_c \leftarrow \{i \mid y_i = c\}, \quad Q_c \leftarrow \{j \mid \ell_j = c\}$

      if $|\mathcal{B}_c| + |Q_c| > 0$ **then**

        $v_c \leftarrow \sum_{i \in \mathcal{B}_c} \tilde{z}_i + \sum_{j \in Q_c} \text{sg}(q_j), \quad \mu_c \leftarrow \frac{v_c}{\|v_c\|_2 + \epsilon}$

      end if

    end for

    $\mathcal{L}_{\text{proto}} \leftarrow \frac{1}{C(C-1)} \sum_{r \neq s} \left(1 + \frac{\|\mu_r - \mu_s\|_2^2}{\sigma^2}\right)^{-1}$

    Step 5: Joint Optimization

    $\mathcal{L} \leftarrow \mathcal{L}_{\text{WCE}} + \lambda_{\text{knn}}(t)\, \mathcal{L}_{\text{knn}} + \lambda_{\text{proto}}(t)\, \mathcal{L}_{\text{proto}}$

    Update model parameters θ, ϕ via AdamW gradient step on $\nabla_{\theta,\phi} \mathcal{L}$

    Step 6: Dynamic Memory Queue Update

    for $i = 1 \ldots B$ **do** ▷ Refresh queue slots with batch embeddings

      if $\ell_p == y_i$ **then**

        $q_p \leftarrow \gamma q_p + (1-\gamma)\tilde{z}_i$ ▷ EMA update for matching class

      else

        $(q_p, \ell_p) \leftarrow (\tilde{z}_i, y_i)$ ▷ Overwrite stale queue slot

      end if

      $q_p \leftarrow \frac{q_p}{\|q_p\|_2}, \quad p \leftarrow (p \bmod M) + 1$ ▷ Re-normalize and advance pointer

    end for

  end for

end for

4 RESULTS AND DISCUSSION

Comparison with Different Loss Functions: Table 1 compares our method against reweighting and contrastive baselines. Our method achieves the strongest overall and balanced accuracy on both datasets, showing the benefit of combining local and global geometric supervision with class-rebalanced classification. On ISIC 2018, SupCon edges ahead on Macro F1, likely due to its sensitivity to precision/recall swings on the smallest classes (DF, VASC); we thus treat overall and balanced accuracy and macro F1 as our primary metrics. We abbreviate Weighted Cross-Entropy loss as Weighted CE or WCE. All the results are run across 5 seed values.

Ablation Study: Table 5 isolates the contribution of each proposed component on top of weighted cross-entropy. The Weighted KNN loss consistently improves performance over the WCE baseline by encouraging local neighbourhood compactness, while the Cauchy Prototype loss provides a larger individual gain through improved global class separation. Combining both objectives yields the strongest results on both datasets, confirming that the two losses address complementary aspects of imbalance-induced geometric distortion rather than contributing redundant supervision. The t-SNE plots in Figure 2 further reinforce this observation, with each component individually improving the clustering structure, while their combination produces the strongest overall compactness and inter-class separation.

Queue Dynamics and Distribution Fidelity: Table 2 reports the class composition of the queue ($M = 4096$) on ISIC 2019. The queue preserves the training distribution throughout training without artificial quotas, and even the rarest class (DF) retains ≈39 entries, well above $k = 10$. Table 3 compares queue designs: a dynamic queue outperforms a fixed per-class quota (512/class), whose rare-class slots take over two epochs to refresh and therefore hold stale features, and an in-batch variant with no queue.

Table 1: Comparison with different loss functions on ISIC 2018 and ISIC 2019 (mean ± std over 5 runs).

Loss Function	Overall Acc (%)	Bal Acc (%)	Macro F1 (%)
	ISIC-18		
Weighted CE [10]	81.22 ± 1.56	75.44 ± 1.72	73.91 ± 1.81
Focal Loss [13]	82.72 ± 1.67	76.84 ± 1.73	76.05 ± 1.90
LDAM [11]	81.42 ± 1.42	75.42 ± 1.65	74.10 ± 1.68
Balanced Softmax [12]	83.72 ± 1.78	77.54 ± 1.79	72.55 ± 1.78
SupCon [14]	86.21 ± 2.01	78.09 ± 1.96	**78.09 ± 1.98**
PaCo [15]	86.31 ± 1.79	78.43 ± 1.83	77.01 ± 1.80
Ours	**87.41 ± 2.03**	**79.32 ± 1.87**	77.89 ± 1.91
	ISIC-19		
Weighted CE [10]	77.10 ± 1.61	66.74 ± 1.56	65.61 ± 1.63
Focal Loss [13]	76.27 ± 1.51	65.91 ± 1.67	66.19 ± 1.64
LDAM [11]	77.77 ± 1.72	67.76 ± 1.79	64.67 ± 1.78
Balanced Softmax [12]	77.30 ± 1.49	64.36 ± 1.52	64.29 ± 1.61
SupCon [14]	81.46 ± 1.68	70.73 ± 1.91	71.27 ± 1.92
PaCo [15]	81.58 ± 1.89	71.19 ± 2.04	70.01 ± 1.95
Ours	**82.78 ± 1.97**	**71.97 ± 1.93**	**72.40 ± 1.91**

Table 2: Queue composition on ISIC 2019 (mean slots per class, $M = 4096$) over epoch ranges, compared with the dataset class proportions.

Phase	NV	MEL	BCC	BKL	AK	SCC	VASC	DF
Ep 1–20	2069	737	541	422	142	105	40	39
Ep 21–40	2089	726	537	425	140	102	40	38
Ep 41–60	2080	731	537	425	139	103	43	39
Queue (%)	50.8	17.9	13.1	10.4	3.4	2.5	1.0	0.9

Table 3: Queue design ablation on ISIC 2019 validation set.

Queue variant	Bal Acc (%)	Macro F1 (%)
In-batch (no queue)	68.02	67.80
Fixed quota (512/class)	71.84	69.85
Dynamic (ours)	**73.52**	**74.20**

Table 4: Behaviour of $\mathcal{L}_{proto}$ during training on ISIC 2019.

Quantity	Epoch 1	Epoch 60
$\|\nabla_\theta \mathcal{L}_{proto}\|_2$	0.1505	0.0018
Intra-class cosine similarity	0.536	0.834
Phase	$\cos(\nabla\mathcal{L}_{CE}, \nabla\mathcal{L}_{proto})$	
Ep 1–20	+0.18	
Ep 21–40	+0.12	
Ep 41–60	+0.09	

Per class Recall Analysis: Table 6 reveals a clear qualitative pattern in how the feature space reorganizes. Rather than uniformly scaling performance, the gains are concentrated in clinically challenging categories — BCC, BKL, SCC, and VASC all show consistent improvements over both SupCon and PaCo. Crucially, the model

Table 5: Component-wise ablation study on ISIC 2018 and ISIC 2019 (mean ± std over 5 runs).

Method	Overall Acc (%)	Bal Acc (%)	Macro F1 (%)
	ISIC-18		
Weighted CE (WCE)	81.22 ± 1.56	75.44 ± 1.72	73.91 ± 1.81
WCE + KNN	84.50 ± 1.77	77.39 ± 1.89	76.32 ± 1.91
WCE + Proto	86.19 ± 1.63	78.43 ± 1.72	76.59 ± 1.87
WCE + KNN + Proto	**87.41 ± 2.03**	**79.32 ± 1.87**	**77.89 ± 1.91**
	ISIC-19		
Weighted CE (WCE)	77.10 ± 1.61	66.74 ± 1.56	65.61 ± 1.63
WCE + KNN	79.87 ± 1.79	70.35 ± 1.81	69.73 ± 1.83
WCE + Proto	81.52 ± 1.71	70.91 ± 1.68	70.74 ± 1.73
WCE + KNN + Proto	**82.78 ± 1.97**	**71.97 ± 1.93**	**72.40 ± 1.91**

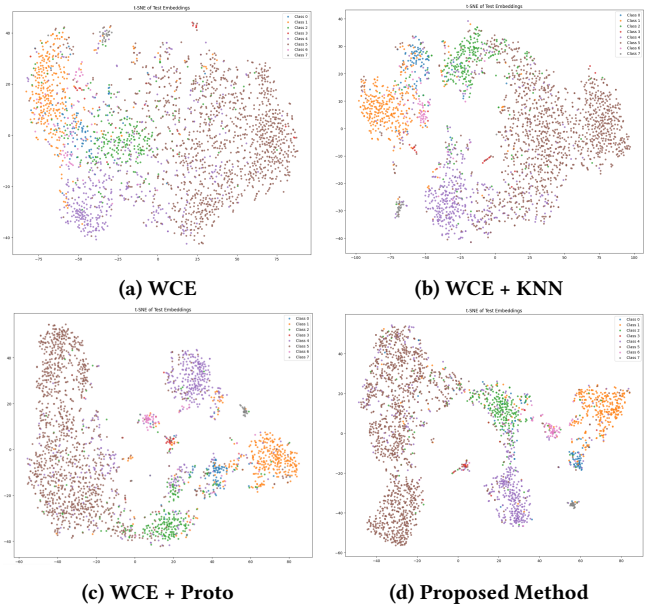
Table 6: Per-class Recall (%), Balanced Accuracy, and Overall Accuracy on ISIC 2019 (mean ± std over 5 runs).

Class	SupCon	PaCo	Ours
	ISIC-19		
NV	91.00 ± 0.28	90.35 ± 0.31	**92.20 ± 0.25**
MEL	77.02 ± 1.22	**79.01 ± 1.34**	77.61 ± 1.15
BCC	74.12 ± 1.45	74.67 ± 1.51	**76.60 ± 1.39**
BKL	65.40 ± 1.61	65.97 ± 1.47	**67.80 ± 1.52**
AK	**52.32 ± 3.15**	51.48 ± 3.01	52.05 ± 2.85
SCC	78.25 ± 3.52	76.41 ± 3.69	**78.95 ± 3.27**
VASC	57.63 ± 5.18	58.81 ± 4.54	**59.79 ± 4.89**
DF	70.10 ± 5.69	**72.82 ± 6.17**	70.76 ± 5.42
Balanced Acc.	70.73 ± 1.91	71.19 ± 2.04	**71.97 ± 1.93**
Overall Acc.	81.46 ± 1.68	81.58 ± 1.89	**82.78 ± 1.97**

maintains high sensitivity on the dominant NV class without allowing it to overwhelm the representations of rarer categories. While PaCo slightly edges ahead on MEL and DF, and SupCon remains competitive on AK, our approach delivers a much more dependable diagnostic balance across the broader lesion spectrum. This behavior reflects the core intent of the dual-scale objective: local compactness anchors fine-grained decision boundaries for scarce samples, while global prototype repulsion keeps rare classes from being subsumed into neighboring clusters—ensuring sensitivity to rare pathologies without sacrificing accuracy on common lesions.

Embedding Quality Analysis: The six metrics in Table 7 jointly evaluate the quality and separability of the learned feature space: **Silhouette** measures class compactness and separation, while **KNN Purity** reflects local neighborhood consistency. **Intra-Class Cosine Similarity** measures within-class feature similarity, whereas **Inter-Class Cosine Similarity** measures similarity between different classes, with lower values indicating better separation. **Margin** captures the gap between intra- and inter-class similarity, and **Separation Ratio** summarizes their relative strength, with higher values indicating better-separated representations. Together, these metrics complement classification accuracy by evaluating **local structure and global class separability**. Table 7 probes the geometric claims directly. Our method achieves the best KNN purity,

inter-class cosine similarity, margin, and separation ratio on both datasets — exactly what $\mathcal{L}_{knn}$ and $\mathcal{L}_{proto}$ are designed to enforce, at the local and global level respectively. SupCon attains marginally higher Silhouette and Intra-Class Cosine Similarity scores on both datasets, yet trails on every other metric, including downstream accuracy. This is not surprising: both Silhouette and Intra-Class Cosine reward tight, majority-dominated clusters and can be inflated by collapsing minority-class embeddings inward without meaningfully separating them from neighbouring classes, whereas our objectives explicitly target inter-class margins for minority classes rather than raw intra-class compactness. Margin and inter-class cosine similarity, both of which improve substantially under our method, track the downstream accuracy gains far more closely — suggesting the improvements reflect genuine geometric restructuring rather than reweighting alone. The t-SNE plots in Figure 3 back this up, with minority-class clusters visibly tighter and better separated under our method.



(a) WCE (b) WCE + KNN

(c) WCE + Proto (d) Proposed Method

Figure 2: t-SNE visualisations of test embeddings on ISIC 2019 across the individual components of the proposed loss function.

Impact of Prototype Separation Kernels: To understand how the geometry of the embedding space responds to different functional forms of global prototype repulsion in $\mathcal{L}_{proto}$, we evaluate four kernel variants $\Phi(r, s)$ operating on pairwise normalized class centroids μ_r, μ_s (Table 8).

The linear kernel exhibits the weakest performance because unweighted dot-product penalization lacks non-linear distance scaling—it penalizes already distant clusters uniformly while failing to deliver focused repulsive force where class overlap is most severe. Introducing a Gaussian (RBF) kernel provides a noticeable improvement over the linear baseline by localizing the repulsion. However, its effectiveness is bounded by steep exponential gradient decay ($\frac{\partial \Phi}{\partial d^2} \propto \exp(-d^2/2\sigma^2)$): once minority-class prototypes achieve even moderate distance from majority centroids, the repulsive force drops off precipitously, leaving rare classes trapped in sub-optimal, partially entangled configurations late in training.

Heavy-tailed formulations resolve this apparent gradient starvation through polynomial rather than exponential decay. Transitioning to the Student-t kernel ($\nu = 3$) yields steady performance gains by maintaining longer-range gradient pressure. Our Cauchy kernel ($\nu = 1$) achieves the strongest overall performance across both benchmarks, delivering substantial relative gains in balanced accuracy and macro metrics over both Gaussian and linear alternatives. The heavy power-law tail ($O(d^{-2})$) of the Cauchy kernel ensures a persistent, non-vanishing repulsive force throughout optimization, continuously driving minority prototypes toward maximal, uniform separation across the unit hypersphere.

Table 7: Embedding quality metrics on ISIC 2018 and ISIC 2019 (mean ± std over 5 runs).

Dataset	Metric	Ours	SupCon	PaCo
ISIC-18	Silhouette (↑)	0.1819 ± 0.0130	**0.1823 ± 0.0211**	0.1367 ± 0.0230
	KNN Purity (↑)	**0.8187 ± 0.0381**	0.7927 ± 0.0397	0.8075 ± 0.0370
	Intra-Class Cos. (↑)	0.7660 ± 0.0315	**0.8656 ± 0.0438**	0.7931 ± 0.0478
	Inter-Class Cos. (↓)	**0.5305 ± 0.0361**	0.6850 ± 0.0431	0.5862 ± 0.0390
	Margin (↑)	**0.2355 ± 0.0231**	0.1806 ± 0.0192	0.2069 ± 0.0199
	Separation Ratio (↑)	**1.44 ± 0.14**	1.26 ± 0.14	1.35 ± 0.15
ISIC-19	Silhouette (↑)	0.2363 ± 0.0170	**0.2484 ± 0.0196**	0.2229 ± 0.0183
	KNN Purity (↑)	**0.7684 ± 0.0327**	0.7209 ± 0.0398	0.7465 ± 0.0361
	Intra-Class Cos. (↑)	0.7878 ± 0.0387	**0.9099 ± 0.0512**	0.8583 ± 0.0400
	Inter-Class Cos. (↓)	**0.5201 ± 0.0304**	0.7816 ± 0.0298	0.6760 ± 0.0412
	Margin (↑)	**0.2677 ± 0.0201**	0.1283 ± 0.0189	0.1823 ± 0.0196
	Separation Ratio (↑)	**1.51 ± 0.17**	1.16 ± 0.13	1.27 ± 0.15

Table 8: Comparative results comparing different kernel functions for prototype separation ($\mathcal{L}_{proto}$) on ISIC 2018 and ISIC 2019 (mean ± std over 5 runs).

Kernel Type	Formulation $\Phi(r, s)$	Overall Acc (%)	Bal Acc (%)	Macro F1 (%)
ISIC-18				
Linear [22]	$\langle \mu_r, \mu_s \rangle$	84.62 ± 1.55	76.15 ± 1.68	75.20 ± 1.62
Gaussian / RBF [20]	$\exp\left(-\frac{\|\mu_r-\mu_s\|_2^2}{2\sigma^2}\right)$	85.80 ± 1.48	77.84 ± 1.56	76.12 ± 1.55
Student-t [23] ($\nu = 3$)	$\left(1+\frac{\|\mu_r-\mu_s\|_2^2}{3\sigma^2}\right)^{-2}$	86.75 ± 1.78	78.62 ± 1.89	77.05 ± 1.85
Cauchy [21] (Ours, $\nu = 1$)	$\left(1+\frac{\|\mu_r-\mu_s\|_2^2}{\sigma^2}\right)^{-1}$	**87.41 ± 2.03**	**79.32 ± 1.87**	**77.89 ± 1.91**
ISIC-19				
Linear [22]	$\langle \mu_r, \mu_s \rangle$	79.25 ± 1.47	68.40 ± 1.41	67.85 ± 1.46
Gaussian / RBF [20]	$\exp\left(-\frac{\|\mu_r-\mu_s\|_2^2}{2\sigma^2}\right)$	81.10 ± 1.51	69.85 ± 1.39	69.90 ± 1.38
Student-t [23] ($\nu = 3$)	$\left(1+\frac{\|\mu_r-\mu_s\|_2^2}{3\sigma^2}\right)^{-2}$	82.12 ± 1.68	71.18 ± 1.83	71.55 ± 1.79
Cauchy [21] (Ours, $\nu = 1$)	$\left(1+\frac{\|\mu_r-\mu_s\|_2^2}{\sigma^2}\right)^{-1}$	**82.78 ± 1.97**	**71.97 ± 1.93**	**72.40 ± 1.91**

Discussion: The results demonstrate the benefit of jointly modeling local and global structure in the embedding space. While the Weighted KNN and Cauchy Prototype losses each improve performance independently, their combination consistently provides further gains across both datasets, indicating that local compactness and global prototype separation address complementary aspects of imbalance-induced feature distortion. Improvements in balanced accuracy and minority-class recall further suggest reduced bias towards majority classes.



(a) SupCon (b) PaCo (c) Proposed Framework

Figure 3: t-SNE visualisations comparing test set embeddings on ISIC 2019 across SupCon, PaCo, and our proposed framework.

The embedding analysis supports this finding, with the proposed method achieving higher KNN purity, lower inter-class similarity, larger feature margins, and stronger separation ratios than the compared methods. Overall, the results suggest that jointly enforcing local and global geometric constraints can yield more discriminative and balanced representations under severe class imbalance.

5 CONCLUSION

We presented a memory-queue-based framework for imbalanced skin lesion classification that jointly targets local feature compactness and global class separation. The method pairs a weighted cross-entropy loss with a weighted KNN loss to improve local neighborhood consistency, alongside a Cauchy Prototype Separation loss that pushes class prototypes further apart globally. An EMA-updated memory queue supplies stable cross-batch representations for both objectives, and a warm-up schedule gradually phases in their contributions as training progresses. Built on a pretrained ResNet-18 backbone, the framework was evaluated on the highly imbalanced ISIC 2018 and ISIC 2019 datasets.

The proposed approach delivers the consistent overall and balanced accuracy among all methods evaluated on both benchmarks. Ablation experiments show that the two geometric objectives complement one another, and embedding analysis reveals gains in neighborhood purity, inter-class separation, feature margins, and overall representation quality. The Cauchy kernel also consistently outperforms the other prototype-separation kernels we tested.

That said, the current study has some limitations: it is evaluated on only two benchmark datasets, relies on a single backbone architecture, and incurs additional computational and memory overhead from the feature queue. Future work will look at stronger architectures, larger and more diverse datasets, adaptive queue management, and dynamically optimized loss and kernel parameters. Cross-dataset evaluation and uncertainty-aware analysis are also planned to further test the framework's robustness and real-world applicability.

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